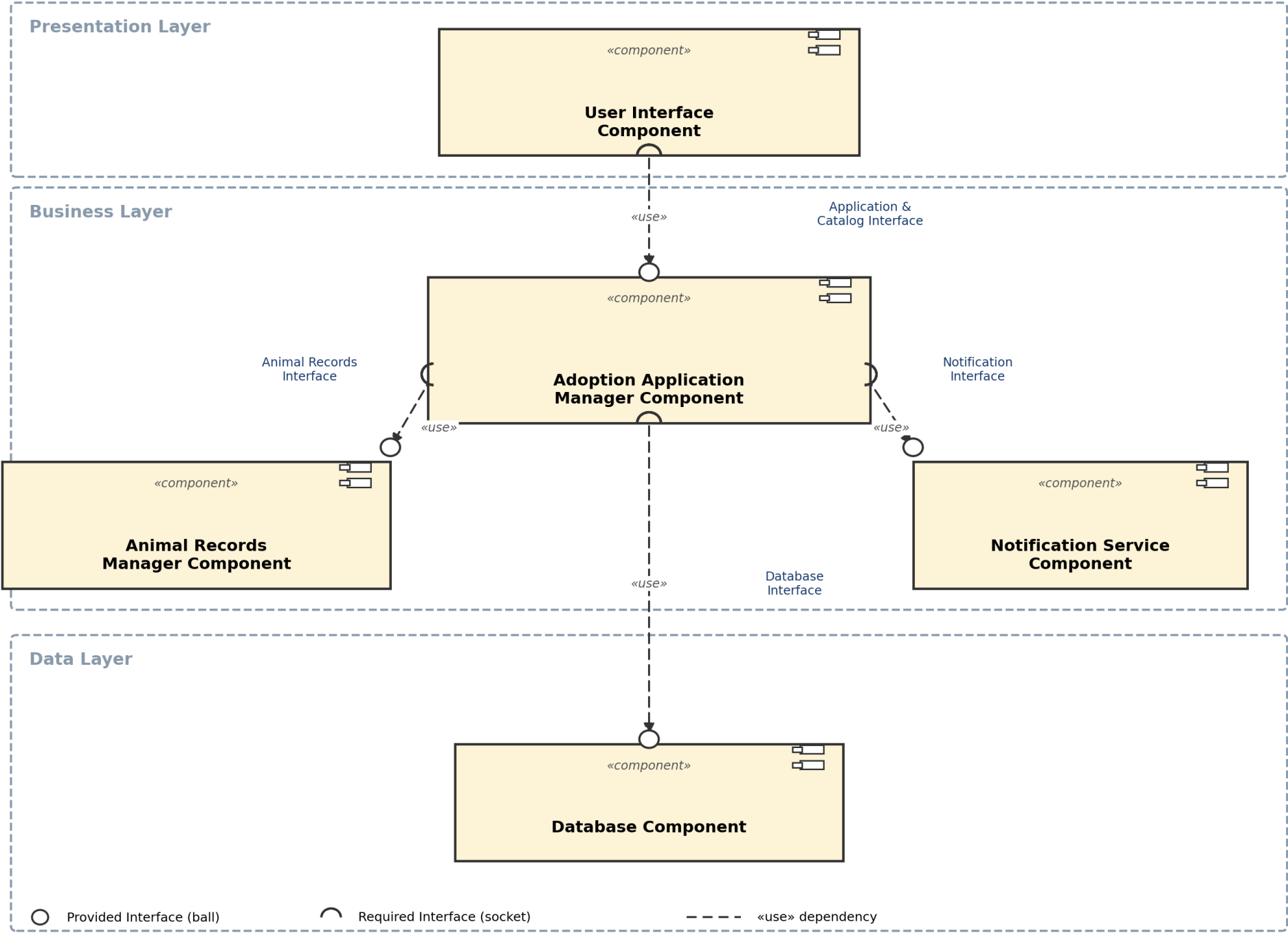


Pet Adoption & Shelter Management System - Layered Architecture



Lab 3: Component Modelling & Architectural Pattern Selection

Architectural Justification

System: Pet Adoption & Shelter Management System (Problem Statement #57)

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Architecture Selection

We chose Layered Architecture for the Pet Adoption & Shelter Management System.

Two Reasons

1. Natural separation of concerns across three actors: Adoption Applicants, Shelter Staff, and Veterinarians all interact with the same underlying workflow (search catalog → submit application → review → approve/reject, plus medical intake and foster placement) but through different screens and permissions. A Presentation layer handles these role-specific views, a Business layer centralizes the shared workflow logic (UC-02 through UC-07, including the UC-05 availability check and the UC-06 home-visit extension), and a Data layer persists animal, application, medical, and foster records. This keeps the review and status-update logic in one place instead of duplicating it per actor.

2. Scale matches a single-shelter deployment: The system serves one shelter (or a small shelter network) with moderate, fairly predictable traffic, rather than requiring independent, elastic scaling of many separate services. A layered design gives centralized control over sensitive workflows — such as who can change an animal's status from FR-001 and FR-003 — and a simpler deployment than a microservices split would need for a system of this size.

Security Advantage

Routing every request through the Business layer — specifically the Adoption Application Manager — before it reaches the Database layer creates a single, consistent checkpoint for the role-based access control required by NFR-002. An Adoption Applicant's session can be restricted to their own application data, while medical and foster records stay reachable only to Shelter Staff and Veterinarians, with encryption at rest and in transit enforced at the Database layer rather than being re-implemented separately for each type of client.

Performance Benefit

Because the layers are deployed together and communicate through direct, in-process calls rather than network calls between independently deployed services, the multi-attribute pet catalog search (Species, Age, Breed, Good-with-Kids) can meet the sub-200ms response time required by NFR-001 without added inter-service network latency. This also keeps the UC-04 review flow — which combines an availability check, a status update, and a notification in one operation — fast and consistent for Shelter Staff.